# DATASET SELECTION & EXPLORATORY ANALYSIS

Milestone 2 | Ozan Polat & Baris Surmelioglu | 9 November 2026

## 1. Research objective

We profile two real social networks and three synthetic controls as groundwork for later graph-to-text encoding comparisons. This milestone reports EDA, not LLM performance.

## 2. Data sources

| Network | Nodes | Edges | Orientation |
|---|---|---|---|
| Facebook (SNAP) | 4,039 | 88,234 | Undirected |
| Email-Eu-core (SNAP) | 1,005 | 25,571 | Directed |
| ER synthetic (n=300) | 300 | 1,613 | Undirected |
| BA synthetic (n=300) | 300 | 891 | Undirected |
| SBM synthetic (n=300) | 300 | 2,290 | Undirected |

Real datasets: Stanford SNAP (Facebook friendships and European institution email ties).

## 3. Exploratory network measurements

| Network | Mean total degree | Density | Components |
|---|---|---|---|
| Facebook | 43.691 | 0.01082 | 1 |
| Email-Eu-core | 50.888 | 0.02534 | 20 (weak) |
| ER | 10.753 | 0.03596 | 1 |
| BA | 5.940 | 0.01987 | 1 |
| SBM | 15.267 | 0.05106 | 1 |

## 4. Observations

Facebook: 4,039 nodes, 88,234 friendships; average total degree 43.7.

Email-Eu-core: 1,005 nodes, 25,571 directed ties; 20 weak and 203 strong components.

Email self-loops: 642, retained for raw-data EDA; 42 anonymized departments.

Within-department directed ties: 36.3% (including self-loops).

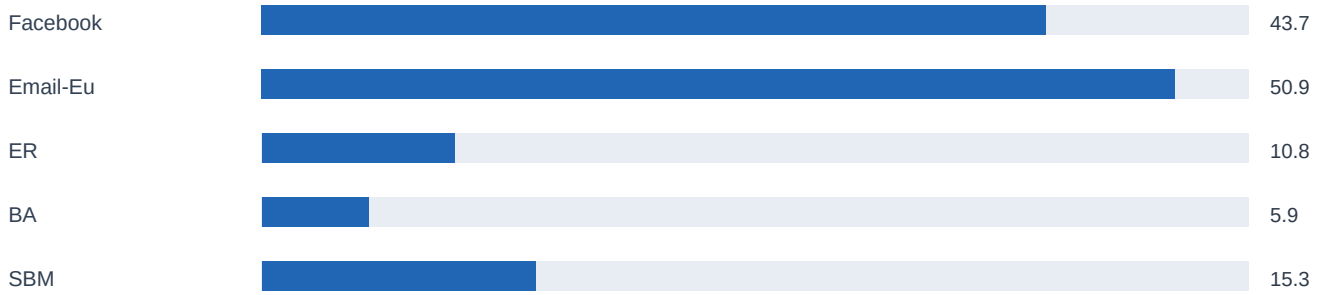
## 5. What the numbers mean

The two SNAP graphs are substantially larger than synthetic 300-node controls.
Density measures the share of possible edges present; edge direction matters.
Reported total degree in email adds incoming and outgoing directed links.

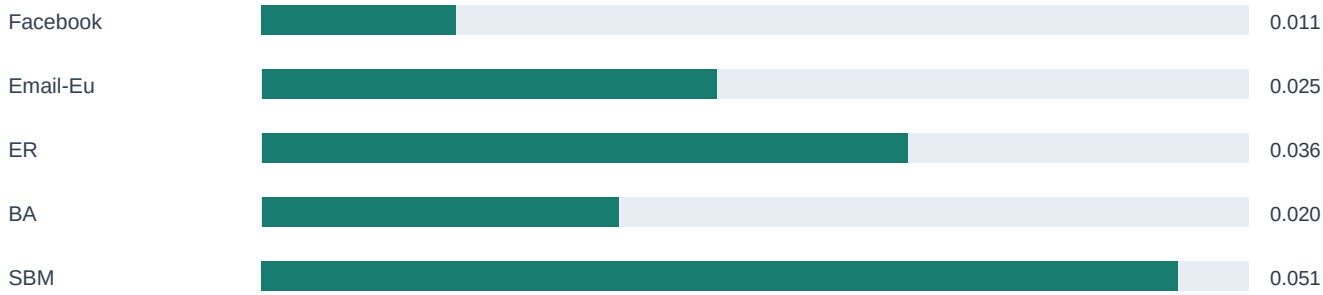
Sources: snap.stanford.edu/data/ego-Facebook.html; snap.stanford.edu/data/email-Eu-core.html

## 6. Observed graph metrics (five datasets)

### A. Average total degree




### B. Graph density




These bars show observed M2 output, not controlled accuracy comparisons.

## 7. Sampling and preparation for later milestones

60 fixed-seed subgraphs (12, 25 and 50 nodes) and 60 reference graph questions were prepared for future edge-list, adjacency-list and natural-language tests. Email direction and department labels are preserved. Self-loops are excluded only from selected later neighbor questions, not from descriptive network statistics.

## 8. Limitations and next steps

Graph sizes, edge direction and densities differ; interpretations are descriptive. Local-neighborhood sampling may over-represent well-connected areas. Future tests should vary random seeds, graph sizes and question types to test robustness.

## Reproducibility and references

Code: github.com/Akephalos7/talk-like-a-graph-swsn

Run: python run_m2.py | Notebook: notebooks/M2_Exploratory_Data_Analysis.ipynb

McAuley & Leskovec (NeurIPS 2012); Yin et al. (KDD 2017); Fatemi et al. (ICLR 2024).

Original SNAP data downloaded from Stanford; not redistributed in this repository.